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Assignment 1

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Quiz questions

1. By convention what is the charge of glass and amber?
2. Why do conductors conduct electricity better than insulators?
3. The electric field between two charged plates is $E = \sigma/(2\epsilon_0)$. True or false?
4. Which of the following statements are correct regarding Gauss' Law (Select all that apply)
   (a) Gauss's Law states that the net electric flux through any closed surface is equal to the enclosed electric charge divided by the permittivity of the medium.
   (b) Gauss's Law applies only to isolated point charges.
   (c) Gauss's Law can be used to determine the electric field produced by continuous charge distributions, such as charged wires or planes.
   (d) Gauss's Law is a consequence of Coulomb's Law.
   (e) Gauss's Law holds true for any shape of the closed surface.
   (f) Gauss's Law is applicable to situations involving both electric and magnetic fields.
5. What are the three types of electrostatic charging?

Quiz answers

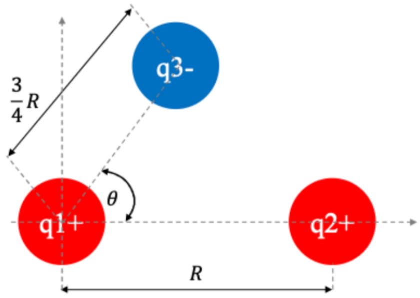
1. Following Benjamin Franklins convention, glass becomes positively charged and amber becomes negatively charged, when charged. This charge results from the triboelectric effect and is therefore not present before rubbing the materials.
2. Conductors conduct electricity as they have mobile electrons (i.e. no real band gap for metals). For insulators the opposite is the case – these have no mobile electrons and can therefore not carry a current (i.e. a large band gap between the valence and conduction bands)
3. False, the electric field between two plates with charge density σ is $E = \sigma/\epsilon_0$. $E = \sigma/(2\epsilon_0)$ is the formula for the electric field due to an infinite sheet.
4. (a) False, Gauss' law states that net electric flux through any closed surface equals the enclosed charge divided by the permittivity of free space ϵ_0.
   (b) False, Gauss' law applies for all charge configurations
   (c) True
   (d) True,
   $$\Phi = \frac{q}{\epsilon_0}$$
   can be directly calculated using the flux integral (assuming rotational symmetry about the middle of ones conductor)
   $$\Phi = \oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{A}$$
   and Coulomb's law
   $$E = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}.$$
   (e) True, as long as the surface is closed.
   (f) False, Gauss' law only concerns itself with electric fields. No contribution from the magnetic field is present anywhere. For magnetism Gauss' law sounds
   $$\oint \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{A} = 0$$

i.e. the magnetic flux through a closed surface is zero (law of no magnetic monopoles).

5. The three types are:
   (a) Charging by friction (Triboelectric effect)
   (b) Charging by contact
   (c) Charging by induction

Problem 1

Consider $q_1 = 1{,}6 \cdot 10^{-19}\,\mathrm{C}$, $q_2 = 3{,}2 \cdot 10^{-19}\,\mathrm{C}$, and $q_3 = -3{,}2 \cdot 10^{-19}\,\mathrm{C}$ with distance $R = 0{,}02\,\mathrm{m}$ and $\theta = 60°$. What is the net electrostatic force on particle 1 from particles 2 and 3?



Givens

$q_1 = 1{,}6 \cdot 10^{-19}\,\mathrm{C}$	Charge of particle 1
$q_2 = 3{,}2 \cdot 10^{-19}\,\mathrm{C}$	Charge of particle 2
$q_3 = -3{,}2 \cdot 10^{-19}\,\mathrm{C}$	Charge of particle 3
$R = 0{,}02\,\mathrm{m}$	Distance between particles 1 and 2
$\theta = 60°$	Angle between charges 2 and 3 measured from charge 1

Derivation

Coulomb's law states that the electrostatic force $\boldsymbol{F}_{12}$ between two point charges q_1 and q_2 separated by $\boldsymbol{r}_{12}$ is

$$\boldsymbol{F}_{12} = k\frac{q_1 q_2}{r_{12}^2}\hat{r}_{12}.$$

We first find the unit vectors $\hat{r}_{12}$ and $\hat{r}_{13}$ as

$$\begin{aligned}\hat{r}_{12} &= -1\hat{\imath}\\ \hat{r}_{13} &= -\cos 60°\,\hat{\imath} - \sin 60°\,\hat{\jmath} = -0{,}5\hat{\imath} - 0{,}866\hat{\jmath}.\end{aligned}$$

This means that the force on particle 1 from particle 2 and 3 is:

$$\begin{aligned}\boldsymbol{F}_{12} &= 8{,}99 \cdot 10^9\,\mathrm{N\,m^2/C^2} \cdot \frac{1{,}6 \cdot 10^{-19}\,\mathrm{C} \cdot 3{,}2 \cdot 10^{-19}\,\mathrm{C}}{(0{,}2\,\mathrm{m})^2} \cdot (-\hat{i})\\ &= -1{,}15 \cdot 10^{-24}\,\mathrm{N}\hat{\imath}\\ \boldsymbol{F}_{13} &= 8{,}99 \cdot 10^9\,\mathrm{N\,m^2/C^2} \cdot \frac{1{,}6 \cdot 10^{-19}\,\mathrm{C} \cdot \left(-3{,}2 \cdot 10^{-19}\,\mathrm{C}\right)}{(0{,}015\,\mathrm{m})^2} \cdot \left(-0{,}5\hat{\imath} - 0{,}866\hat{\jmath}\right)\\ &= -2{,}05 \cdot 10^{-24}\,\mathrm{N}\left(-0{,}5\hat{\imath} - 0{,}866\hat{\jmath}\right)\\ &= 1{,}02 \cdot 10^{-24}\,\mathrm{N}\hat{\imath} + 1{,}77 \cdot 10^{-24}\,\mathrm{N}\hat{\jmath}.\end{aligned}$$

Using the principle of superposition the total force on particle 1 is then:

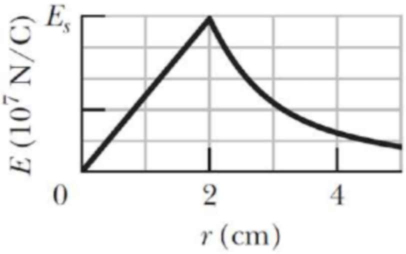
$$\boldsymbol{F}_{1,\mathrm{net}} = \boldsymbol{F}_{12} + \boldsymbol{F}_{13} = \left(1{,}02 \cdot 10^{-24}\,\mathrm{N} - 1{,}15 \cdot 10^{-24}\,\mathrm{N}\right)\hat{\imath} + 1{,}77 \cdot 10^{-24}\,\mathrm{N}\hat{\jmath} = -0{,}13 \cdot 10^{-24}\,\mathrm{N}\hat{\imath} + 1{,}77 \cdot 10^{-24}\,\mathrm{N}\hat{\jmath}.$$

$$\boxed{\boldsymbol{F}_{1,\text{net}} = -0{,}13 \cdot 10^{-24}\,\text{N}\,\hat{\imath} + 1{,}77 \cdot 10^{-24}\,\text{N}\,\hat{\jmath}}$$ RESULT

Problem 2

The figure on the right, gives the magnitude of the electric field inside and outside a sphere with a positive charge distributed uniformly throughout its volume. The scale of the vertical axis is set by $E_s = 10 \cdot 10^7 \, \frac{\mathrm{N}}{\mathrm{C}}$.

1. What is the charge on the sphere?
2. What is the field magnitude at $r = 8{,}0\,\mathrm{m}$?



Givens

$$\begin{array}{r|l} E_s = 10 \cdot 10^7 \, \frac{\mathrm{N}}{\mathrm{C}} & \text{Scale of vertical axis} \\ R = 2\,\mathrm{cm} & \text{Radius of sphere} \end{array}$$

Derivation

As charge exists in the sphere the sphere itself must be insulating (no charge exists within a conducting sphere) from the figure we see that the radius of the sphere is 2 cm . Just on the edge (or outside) of a uniformly charged sphere, the field will be as if the sphere was a point charge:

$$E_s = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^2}.$$

Solving for the charge Q we obtain

$$Q_s = 4\pi\epsilon_0 R^2 E_s = 4\pi \cdot 8{,}85 \cdot 10^{-12}\,\mathrm{C}^2/\mathrm{Nm}^2 \cdot (2\,\mathrm{cm})^2 \cdot 10 \cdot 10^7 \,\frac{\mathrm{N}}{\mathrm{C}} = 4{,}4 \cdot 10^{-7}\,\mathrm{C}.$$

At $r = 8{,}0\,\mathrm{m}$ we're outside of the sphere, where the E-field is:

$$E(8{,}0\,\mathrm{m}) = \frac{1}{4\pi\epsilon_0} \frac{Q_s}{r^2}.$$

Plugging in all the known values we get:

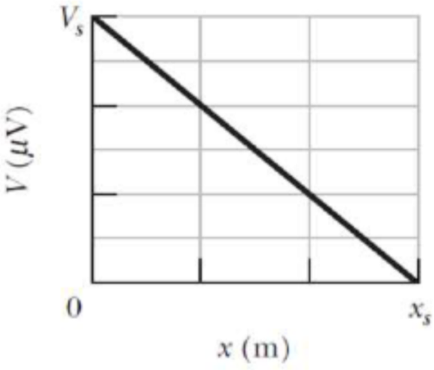
$$E(8{,}0\,\mathrm{m}) = \frac{1}{4\pi \cdot 8{,}85 \cdot 10^{-12}\,\mathrm{C}^2/\mathrm{Nm}^2} \cdot \frac{4{,}4 \cdot 10^{-7}\,\mathrm{C}}{(8{,}0\,\mathrm{m})^2} = 61{,}8\,\frac{\mathrm{N}}{\mathrm{C}}.$$

$$\boxed{\begin{aligned} Q_s &= 4{,}4 \cdot 10^{-7}\,\mathrm{C} \\ E(8{,}0\,\mathrm{m}) &= 61{,}8\,\frac{\mathrm{N}}{\mathrm{C}}. \end{aligned}}$$

RESULT

Problem 3

The figure on the right gives the electric potential $V(x)$ along a copper wire carrying a uniform current, from a point of higher potential, $V_s = 12\,\mu\text{V}$ at $x = 0$ to a point of zero potential at $x_s = 3\,\text{m}$. The wire has a radius of 2,20 mm. What is the current in the wire?
Hint: Use the resistivity of the copper $\rho_{\text{Cu}} = 1{,}68 \cdot 10^{-8}\,\Omega\,\text{m}$



Givens

$$\begin{array}{r|l} V_s = 12\,\mu\text{V} & \text{Potential at } x = 0 \\ x_s = 3\,\text{m} & \text{Distance for } V = 0 \\ r = 2{,}20\,\text{mm} & \text{Radius of wire} \\ \rho_{\text{cu}} = 1{,}68 \cdot 10^{-8}\,\Omega\,\text{m} & \text{Resistivity of copper} \end{array}$$

Derivation

Resistivity is defined as:

$$\rho = \frac{RA}{l}$$

where R is resistance, A is cross-sectional area and l is length. Solving this for the resistance R we obtain:

$$R = \frac{\rho_{\text{cu}} x_s}{\pi \cdot r^2}.$$

Plugging in known values we get:

$$R = \frac{1{,}68 \cdot 10^{-8}\,\Omega\,\text{m} \cdot 3\,\text{m}}{\pi \cdot (2{,}20\,\text{mm})^2} = 3{,}3 \cdot 10^{-3}\,\Omega.$$

Ohm's law states that the current is given by the ratio of the voltage difference and the resistance:

$$I = \frac{U}{R}.$$

Which gives:

$$I = \frac{V_s}{R} = \frac{12\,\mu\text{V}}{3{,}3 \cdot 10^{-3}\,\Omega} = 3{,}64\,\text{mA}.$$

$$\boxed{I = 3{,}64\,\text{mA}}$$ RESULT